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RoPE (RoFormer)

原始文献：RoFormer：Enhanced Transformer with Rotary Position Embedding, Jianlin Su, arXiv:2104.09864v5

Introduction

In this paper, we introduce a novel method, namely **Rotary Position Embedding**(RoPE), to leverage the positional information into the learning process of PLMS(pre-trained language models).

旋转位置嵌入

Specifically, RoPE encodes the **absolute position** with a **rotation matrix** and meanwhile incorporates the **explicit relative position dependency** in self-attention formulation.

RoPE 用旋转矩阵编码绝对位置，同时将显式的相对位置依赖信息融合于自注意力计算。

Note that the proposed RoPE is prioritized over the existing methods through valuable properties, including the sequence length flexibility, decaying inter-token dependency with increasing relative distances, and the capability of equipping the linear self-attention with relative position encoding.

序列长度的灵活性，相对位置增大时 token 间的依赖性衰减，

Experimental results on various long text classification benchmark datasets show that the enhanced transformer with rotary position embedding, namely RoFormer, can give better performance compared to baseline alternatives and thus demonstrates the efficacy of the proposed RoPE.

In brief, our contributions are three-folds as follows:

(1) We investigated the existing approaches to the relative position encoding and found that they are mostly built based on the idea of the decomposition of **adding position encoding** to the **context representations**.

现有的相对位置编码方法，上下文表达叠加位置编码。

We introduce a novel method, namely Rotary Position Embedding(RoPE), to leverage the positional information into the learning process of PLMS.

The key idea is to encode relative position by multiplying the **context representations** with a **rotation matrix** with a clear theoretical interpretation.

新的方法，将上下文表达乘以一个旋转矩阵。

(2) We study the properties of RoPE and show that it decays with the relative distance increased, which is desired for natural language encoding.

We kindly argue that previous relative position encoding-based approaches are not compatible with linear self-attention.

(3) We evaluate the proposed RoFormer on various long text benchmark datasets.

Our experiments show that it consistently achieves better performance compared to its alternatives.

Some experiments with pre-trained language models are available on GitHub:

<https://github.com/ZhuiyiTechnology/roformer>.

(end of three-folds)

The remaining of the paper is organized as follows.

We establish a formal description of the position encoding problem in self-attention architecture and revisit previous works in Section (2).

We then describe the rotary position encoding(RoPE) and study its properties in Section (3).

We report experiments in Section (4).

Finally, we conclude this paper in Section (5).

2 Background and Related Work

2.1 Preliminary

Let $\mathbb{S}_N = \{w_i\}_{i=1}^N$ be a sequence of N input tokens with w_i being the i^{th} element.
 $\mathbb{S}_N$ 是 token 序列, w_i 是 token

The corresponding word embedding of $\mathbb{S}_N$ is denoted as $\mathbb{E}_N = \{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the d -dimensional word embedding vector of token w_i without position information.
 $\mathbb{E}_N$ 是嵌入向量序列, $\mathbf{x}_i$ 是 w_i 的 d -维词嵌入向量, 不含位置信息

The self-attention first incorporates position information to the word embeddings and transforms them into queries, keys, and value representations.

$$\begin{aligned}\mathbf{q}_m &= f_q(\mathbf{x}_m, m) \\ \mathbf{k}_n &= f_k(\mathbf{x}_n, n) \\ \mathbf{v}_n &= f_v(\mathbf{x}_n, n)\end{aligned}$$

(1)

where $\mathbf{q}_m$, $\mathbf{k}_n$ and $\mathbf{v}_n$ incorporate the m^{th} and n^{th} positions through f_q , f_k and f_v , respectively.

解释: 这里先形式化地定义位置嵌入的公式, 即嵌入函数 f_q , f_k and f_v , 原文不是直接给出这个函数, 而是先分析这个嵌入函数应该满足什么条件。公式(1)补充如下:

$$\begin{cases} \mathbf{q}_m = f_q(\mathbf{x}_m, m) & m = 1, \dots, N \\ \mathbf{k}_n = f_k(\mathbf{x}_n, n) & n = 1, \dots, N \\ \mathbf{v}_n = f_v(\mathbf{x}_n, n) & n = 1, \dots, N \end{cases}$$

The query and key values are then used to compute the attention weights, while the output is computed as the weighted sum over the value representation.

$$a_{m,n} = \frac{\exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_n}{\sqrt{d}}\right)}{\sum_{j=1}^N \exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_j}{\sqrt{d}}\right)}$$

$$\mathbf{o}_m = \sum_{n=1}^N a_{m,n} \mathbf{v}_n$$

(2)

公式(2)补充，[这里的向量都是竖写格式](#)，先固定一个 m ($m = 1, \dots, N$)，向量 $\bar{\mathbf{a}}_m$ 表示 $\mathbf{q}_m$ 与每个 $\mathbf{k}_n$ 求内积后组成的归一化权值向量，

$$\bar{\mathbf{a}}_m = \begin{pmatrix} a_{m,1} \\ \vdots \\ a_{m,N} \end{pmatrix} \quad \sum_{j=1}^N a_{m,j} = 1 \quad m = 1, \dots, N$$

用权值向量 $\bar{\mathbf{a}}_m$ 与向量组 $\{\mathbf{v}_n\}_{n=1, \dots, N}$ 加权求和得到第 m 个输出 $\mathbf{o}_m$ ，对每个 m 如此计算，得到所有的输出 $\{\mathbf{o}_m\}_{m=1, \dots, N}$ 。

解释：这个公式就是 attention 算法，与 transformer 原文中的公式是一致的。

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V = \text{softmax}_{\text{by_row}}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The existing approaches of transformer-based position encoding mainly focus on choosing a suitable function to form Equation (1).

2.2 Absolute position embedding

绝对位置编码

A typical choice of Equation (1) is

$$f_{t:t \in \{q,k,v\}}(\mathbf{x}_i, i) := \mathbf{W}_{t:t \in \{q,k,v\}}(\mathbf{x}_i + \mathbf{p}_i)$$

(3)

where $\mathbf{p}_i \in \mathbb{R}^d$ is a d-dimensional vector depending of the position of token $\mathbf{x}_i$.

用不含位置信息的向量 $\mathbf{x}_i$ 叠加位置向量 $\mathbf{p}_i$ 得到包含位置信息的新向量 $\mathbf{q}_i$, $\mathbf{k}_i$ 和 $\mathbf{v}_i$ 这样定义的位置向量 $\mathbf{p}_i$ ，其实只取决于位置 i ，所以，可以通过预定义或者预训练得到。

公式(3)的解释，其实是三个公式，

$$f_{t:t \in \{q,k,v\}}(\mathbf{x}_i, i) := \mathbf{W}_{t:t \in \{q,k,v\}}(\mathbf{x}_i + \mathbf{p}_i)$$

$$\begin{cases} f_q(\mathbf{x}_i, i) := \mathbf{W}_q(\mathbf{x}_i + \mathbf{p}_i) \\ f_k(\mathbf{x}_i, i) := \mathbf{W}_k(\mathbf{x}_i + \mathbf{p}_i) \\ f_v(\mathbf{x}_i, i) := \mathbf{W}_v(\mathbf{x}_i + \mathbf{p}_i) \end{cases}$$

其中 $\mathbf{W}_q$, $\mathbf{W}_k$, $\mathbf{W}_v$ ，分别是 Q, K, V 的投影阵，即参数矩阵。

预训练方法

Previous work Devlin et al. [2019], Lan et al. [2020], Clark et al. [2020], Radford et al. [2019], Radford and Narasimhan [2018] introduced the use of a **set of trainable vectors** $\mathbf{p}_i \in \{\mathbf{p}_t\}_{t=1}^L$, where L is the maximum sequence length.

预定义方法

The authors of Vaswani et al. [2017] have proposed to generate $\mathbf{p}_i$ using the sinusoidal function.

$$\begin{cases} \mathbf{p}_{i,2t} = \sin\left(\frac{k}{10000^{\frac{2t}{d}}}\right) \\ \mathbf{p}_{i,2t+1} = \cos\left(\frac{k}{10000^{\frac{2t}{d}}}\right) \end{cases}$$

(4)

in which $\mathbf{p}_{i,2t}$ is the $2t^{th}$ element of the d -dimensional vector $\mathbf{p}_i$.

In the next section, we show that our proposed RoPE is related to this intuition from the sinusoidal function perspective.

However, instead of directly adding the position to the context representation, RoPE proposes to incorporate the relative position information by multiplying with the sinusoidal functions.

2.3 Relative position embedding

相对位置编码

The authors of Shaw et al. [2018] applied different settings of Equation (1) as following:

$$\begin{aligned} f_q(\mathbf{x}_m) &:= \mathbf{W}_q \mathbf{x}_m \\ f_k(\mathbf{x}_n, n) &:= \mathbf{W}_k (\mathbf{x}_n + \tilde{\mathbf{p}}_r^k) \\ f_v(\mathbf{x}_n, n) &:= \mathbf{W}_v (\mathbf{x}_n + \tilde{\mathbf{p}}_r^v) \end{aligned}$$

(5)

where $\tilde{\mathbf{p}}_r^k, \tilde{\mathbf{p}}_r^v \in \mathbb{R}^d$ are trainable relative position embeddings.

Note that $r = \text{clip}(m - n, r_{\min}, r_{\max})$ represents the relative distance between position m and n .

其中的 clip 函数

$$\text{clip}(x, \min, \max) = \begin{cases} \min & x < \min \\ x & \min \leq x \leq \max \\ \max & x > \max \end{cases}$$

They clipped the relative distance with the hypothesis that precise relative position information is not useful beyond a certain distance.

Keeping the form of Equation (3), the authors Dai et al. [2019] have proposed to decompose $\mathbf{q}_m^T \mathbf{k}_n$ of Equation (2) as

$$\mathbf{q}_m^T \mathbf{k}_n = \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{p}_n + \mathbf{p}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + \mathbf{p}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{p}_n$$

(6)

解释:

$$\mathbf{q}_m = \mathbf{W}_q (\mathbf{x}_m + \mathbf{p}_m)$$

$$\mathbf{k}_n = \mathbf{W}_k (\mathbf{x}_n + \mathbf{p}_n)$$

$$\mathbf{q}_m^T \mathbf{k}_n = (\mathbf{W}_q (\mathbf{x}_m + \mathbf{p}_m))^T (\mathbf{W}_k (\mathbf{x}_n + \mathbf{p}_n)) = (\mathbf{x}_m^T \mathbf{W}_q^T + \mathbf{p}_m^T \mathbf{W}_q^T) (\mathbf{W}_k \mathbf{x}_n + \mathbf{W}_k \mathbf{p}_n)$$

展开上式即得到公式(6)

进一步的解释: 公式(6)中的左边 $\mathbf{q}_m^T \mathbf{k}_n$ 是合成相关度 (attention),

右边第一项 $\mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n$ 是非位置信息的嵌入向量 $\mathbf{x}_m$ 和 $\mathbf{x}_n$ 的相关度,

右边第二项 $\mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{p}_n$ 是非位置信息的嵌入向量 $\mathbf{x}_m$ 和位置信息的嵌入向量 $\mathbf{p}_n$ 的相关度,

右边第三项 $\mathbf{p}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n$ 是非位置信息的嵌入向量 $\mathbf{x}_n$ 和位置信息的嵌入向量 $\mathbf{p}_m$ 的相关度,

右边第四项 $\mathbf{p}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{p}_n$ 是位置信息的嵌入向量 $\mathbf{p}_m$ 和 $\mathbf{p}_n$ 的相关度,

在以上拆解的基础上, 可以对每个部分进行改进。

下面对公式(6)进行改进,

the key idea is to replace the absolute position embedding $\mathbf{p}_n$ with its sinusoid-encoded relative counterpart $\tilde{\mathbf{p}}_{m-n}$, while the absolute position $\mathbf{p}_m$ in the third and fourth term with two trainable vectors $\mathbf{u}$ and $\mathbf{v}$ independent of the query positions.

将绝对位置嵌入 $\mathbf{p}_n$ 改为相对位置 $\tilde{\mathbf{p}}_{m-n}$, 并且第 3 项和第 4 项的绝对位置 $\mathbf{p}_m$ 也改为独立于 query 位置的可训练向量 $\mathbf{u}$ 和 $\mathbf{v}$,

Further, $\mathbf{W}_k$ is distinguished for the content-based and location-based key vectors $\mathbf{x}_n$ and $\mathbf{p}_n$, denoted as $\mathbf{W}_k$ and $\tilde{\mathbf{W}}_k$, resulting in:

$$\mathbf{q}_m^T \mathbf{k}_n = \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^T \mathbf{W}_q^T \tilde{\mathbf{W}}_k \tilde{\mathbf{p}}_{m-n} + \mathbf{u}^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + \mathbf{v}^T \mathbf{W}_q^T \tilde{\mathbf{W}}_k \tilde{\mathbf{p}}_{m-n}$$

(7)

解释: $\mathbf{W}_k$ 对 content-based 的 key vectors $\mathbf{x}_n$ 与 location-based 的 $\mathbf{p}_n$ 是不同的, 所以, 将公式(6)中的 $\mathbf{k}_n = \mathbf{W}_k (\mathbf{x}_n + \mathbf{p}_n)$, 改为 $\mathbf{k}_n = \mathbf{W}_k \mathbf{x}_n + \tilde{\mathbf{W}}_k \mathbf{p}_n$, 也就是说, 对 $\mathbf{x}_n$ 和 $\mathbf{p}_n$ 应用不同的投影矩阵。

It is noteworthy that the position information in the value term is removed by setting $f_v(\mathbf{x}_j) = \mathbf{W}_v \mathbf{x}_j$.

解释: 对 value term $\mathbf{v}_j = f_v(\mathbf{x}_j) = \mathbf{W}_v \mathbf{x}_j$, 可以不用叠加位置信息。

Later work Raffel et al. [2020], He et al. [2020], Ke et al. [2020], Huang et al. [2020] followed these settings by only encoding the relative position information into the attention weights.

However, the authors of Raffel et al. [2020] reformed Equation (6) as:

$$\mathbf{q}_m^T \mathbf{k}_n = \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + b_{i,j}$$

(8)

where $b_{i,j}$ is a trainable bias.

The authors of Ke et al. [2020] investigated the middle two terms of Equation (6) and found little correlations between absolute positions and words.

The authors of Raffel et al. [2020] proposed to model a pair of words or positions using different projection matrices.

$$\mathbf{q}_m^T \mathbf{k}_n = \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + \mathbf{p}_m^T \mathbf{U}_q^T \mathbf{U}_k \mathbf{p}_n + b_{i,j}$$

(9)

The authors of He et al. [2020] argued that the relative positions of two tokens could only be fully modeled using the middle two terms of Equation (6).

As a consequence, the absolute position embeddings $\mathbf{p}_m$ and $\mathbf{p}_n$ were simply replaced with the relative position embeddings $\tilde{\mathbf{p}}_{m-n}$

$$\mathbf{q}_m^T \mathbf{k}_n = \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^T \mathbf{W}_q^T \mathbf{W}_k \tilde{\mathbf{p}}_{m-n} + \tilde{\mathbf{p}}_{m-n}^T \mathbf{W}_q^T \mathbf{W}_k \mathbf{x}_n$$

(10)

A comparison of the four variants of the relative position embeddings Radford and Narasimhan [2018] has shown that the variant similar to Equation (10) is the most efficient among the other three.

Generally speaking, all these approaches attempt to **modify Equation (6)** based on the **decomposition of Equation (3)** under the **self-attention settings in Equation (2)**, which was originally proposed in Vaswani et al. [2017].

They commonly introduced to directly **add** the **position information** to the **context representations**.

以上各种方法，都是叠加法，位置向量+上下文向量

Unlikely, our approach aims to derive the **relative position encoding** from Equation (1) under some constraints.

本文方法，从公式(1)出发，在满足一些条件下，导出相对位置编码

Next, we show that the derived approach is more interpretable by incorporating relative position information with the **rotation** of context representations.

对上下文向量进行旋转，融合相对位置信息

3 Proposed approach

In this section, we discuss the proposed **rotary position embedding** (RoPE).

We first formulate the relative position encoding problem in Section (3.1), we then derive the RoPE in Section (3.2) and investigate its properties in Section (3.3).

3.1 Formulation

Transformer-based language modeling usually leverages the position information of individual tokens through a self-attention mechanism.

As can be observed in Equation (2), $\mathbf{q}_m^T \mathbf{k}_n$ typically enables knowledge conveyance between tokens at different positions.

$$a_{m,n} = \frac{\exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_n}{\sqrt{d}}\right)}{\sum_{j=1}^N \exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_j}{\sqrt{d}}\right)} \quad \begin{cases} m = 1, \dots, N \\ n = 1, \dots, N \end{cases} \quad \mathbf{o}_m = \sum_{n=1}^N a_{m,n} \mathbf{v}_n \quad m = 1, \dots, N$$

(2)

In order to incorporate relative position information, we require the inner product of query $\mathbf{q}_m$ and key $\mathbf{k}_n$ to be formulated by a function g , which takes only the word embeddings $\mathbf{x}_m$, $\mathbf{x}_n$, and their **relative position** ($m - n$) as input variables.

In other words, we hope that the **inner product** encodes **position information** only in the relative form:

$$\mathbf{q}_m^T \mathbf{k}_n = \langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, m - n)$$

(11)

The ultimate goal is to find an equivalent encoding mechanism to solve the functions $f_q(\mathbf{x}_m, m)$ and $f_k(\mathbf{x}_n, n)$ to conform the aforementioned relation.

注释: $\langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle$ 为内积计算

3.2 Rotary position embedding

3.2.1 A 2D case

We begin with a simple case with a dimension $d = 2$.

以 2 维为例

Under these settings, we make use of the geometric property of vectors on a 2D plane and its complex form to prove (refer Section (3.4.1) for more details) that a solution to our formulation Equation (11) is:

$$\begin{aligned} f_q(\mathbf{x}_m, m) &= (\mathbf{W}_q \mathbf{x}_m) e^{im\theta} \\ f_k(\mathbf{x}_n, n) &= (\mathbf{W}_k \mathbf{x}_n) e^{in\theta} \\ g(\mathbf{x}_m, \mathbf{x}_n, m - n) &= \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}] \end{aligned}$$

(12)

where $\text{Re}[\cdot]$ is the real part of a complex number and $(\mathbf{W}_k \mathbf{x}_n)^*$ represents the **conjugate complex** number of $(\mathbf{W}_k \mathbf{x}_n)$.

共轭复数

解释: 公式(12)的前两个公式表明, 位置信息嵌入是独立进行的, 即**绝对位置信息嵌入**,

公式 $f_q(\mathbf{x}_m, m) = (\mathbf{W}_q \mathbf{x}_m) e^{im\theta}$ 表示位置 m 的信息嵌入 $\mathbf{x}_m$ 时, 与其他位置 n 无关, 公式 $f_k(\mathbf{x}_n, n) = (\mathbf{W}_k \mathbf{x}_n) e^{in\theta}$ 表示位置 n 的信息嵌入 $\mathbf{x}_n$ 时, 也与其他位置 m 无关, 而且这种嵌入可以拆分为与位置信息无关的投影 $(\mathbf{W}_q \mathbf{x}_m)$ 、 $(\mathbf{W}_k \mathbf{x}_n)$ 以及单纯的**绝对位置信息嵌入**算子 $e^{im\theta}$ 、 $e^{in\theta}$,

公式(12)的第三个公式表明, 嵌入位置信息后, 求相关度 (attention) 的表达式 $g(\mathbf{x}_m, \mathbf{x}_n, m - n)$ 可以拆分为两个与位置信息无关的投影 $(\mathbf{W}_q \mathbf{x}_m)$ 和 $(\mathbf{W}_k \mathbf{x}_n)^*$, 以及单纯的**相对位置信息嵌入**算子 $e^{i(m-n)\theta}$,

$\theta \in \mathbb{R}$ is a preset non-zero constant.

θ 为预设的非零常数

第一步：先解释公式(12)中的前两个， $f_q(\mathbf{x}_m, m) = (\mathbf{W}_q \mathbf{x}_m) e^{im\theta}$ ， $f_k(\mathbf{x}_n, n) = (\mathbf{W}_k \mathbf{x}_n) e^{in\theta}$

解释：先看一个复数 z 旋转 θ 角度例子，假设有复数 $z = a + ib$ ，其对应的向量形式为 $z = \begin{pmatrix} a \\ b \end{pmatrix}$ ，

则复数乘积

$$z \cdot e^{i\theta} = (a + ib) \cdot e^{i\theta} = (a + ib)(\cos\theta + i\sin\theta) = (a \cdot \cos\theta - b \cdot \sin\theta) + i(a \cdot \sin\theta + b \cdot \cos\theta)$$

如果写为矩阵形式

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \cdot \cos\theta - b \cdot \sin\theta \\ a \cdot \sin\theta + b \cdot \cos\theta \end{pmatrix}$$

结论：复数乘法 $(a + ib) \cdot e^{i\theta}$ 与矩阵乘法 $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$ 的结果是等效对应的，称矩阵

$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ 为旋转阵。

We can further write $f_{\{q,k\}}(\mathbf{x}_m, m)$ in a multiplication matrix:

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$$

(13)

where $\begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$ is $\mathbf{x}_m$ expressed in the 2D coordinates.

解释：公式(12)为复数形式，先假设矩阵形式为

$$\begin{aligned} \mathbf{W}_q &= \begin{pmatrix} W_q^{(11)} & W_q^{(12)} \\ W_q^{(21)} & W_q^{(22)} \end{pmatrix} & \mathbf{x}_m &= \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix} & \mathbf{W}_q \mathbf{x}_m &= \begin{pmatrix} W_q^{(11)} x_m^{(1)} + W_q^{(12)} x_m^{(2)} \\ W_q^{(21)} x_m^{(1)} + W_q^{(22)} x_m^{(2)} \end{pmatrix} \\ \mathbf{W}_k &= \begin{pmatrix} W_k^{(11)} & W_k^{(12)} \\ W_k^{(21)} & W_k^{(22)} \end{pmatrix} & \mathbf{x}_n &= \begin{pmatrix} x_n^{(1)} \\ x_n^{(2)} \end{pmatrix} & \mathbf{W}_k \mathbf{x}_n &= \begin{pmatrix} W_k^{(11)} x_n^{(1)} + W_k^{(12)} x_n^{(2)} \\ W_k^{(21)} x_n^{(1)} + W_k^{(22)} x_n^{(2)} \end{pmatrix} \end{aligned}$$

将 $\mathbf{W}_q \mathbf{x}_m$ $\mathbf{W}_k \mathbf{x}_n$ 视为复数（均为二维向量），则公式(12)改写为

$$f_q(\mathbf{x}_m, m) = (\mathbf{W}_q \mathbf{x}_m) e^{im\theta} = \left((W_q^{(11)} x_m^{(1)} + W_q^{(12)} x_m^{(2)}) + i(W_q^{(21)} x_m^{(1)} + W_q^{(22)} x_m^{(2)}) \right) \cdot e^{im\theta}$$

$$f_k(\mathbf{x}_n, n) = (\mathbf{W}_k \mathbf{x}_n) e^{in\theta} = \left((W_k^{(11)} x_n^{(1)} + W_k^{(12)} x_n^{(2)}) + i(W_k^{(21)} x_n^{(1)} + W_k^{(22)} x_n^{(2)}) \right) \cdot e^{in\theta}$$

参考前面复数 z 旋转 θ 角度例子，得到对应的矩阵形式的公式(13)

$$f_q(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_q^{(11)} & W_q^{(12)} \\ W_q^{(21)} & W_q^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix}$$

$$f_k(\mathbf{x}_n, n) = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix} \begin{pmatrix} W_k^{(11)} & W_k^{(12)} \\ W_k^{(21)} & W_k^{(22)} \end{pmatrix} \begin{pmatrix} x_n^{(1)} \\ x_n^{(2)} \end{pmatrix}$$

第二步：再解释公式(12)中的第三个， $g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}]$

假设有向量 $\vec{z}_1 = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$, $\vec{z}_2 = \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$, 向量内积 $\langle \vec{z}_1, \vec{z}_2 \rangle = (a_1 \cdot a_2 + b_1 \cdot b_2)$,

对应的复数表示形式

$$\begin{cases} z_1 = a_1 + i \cdot b_1 \\ z_2 = a_2 + i \cdot b_2 \end{cases}$$

$$z_1 \cdot (z_2)^* = (a_1 + i \cdot b_1) \cdot (a_2 + i \cdot b_2)^*$$

$$z_1 \cdot (z_2)^* = (a_1 + i \cdot b_1) \cdot (a_2 - i \cdot b_2)$$

$$z_1 \cdot (z_2)^* = (a_1 \cdot a_2 + b_1 \cdot b_2) + i \cdot (-a_1 b_2 + a_2 b_1)$$

可以得到结论，向量内积 $\langle \vec{z}_1, \vec{z}_2 \rangle$ 等于对应复数乘积的实部 $\text{Re}[z_1 \cdot (z_2)^*]$

$$\langle \vec{z}_1, \vec{z}_2 \rangle = (a_1 \cdot a_2 + b_1 \cdot b_2) = \text{Re}[z_1 \cdot (z_2)^*]$$

下面推导公式(12)中的第三个 $g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}]$

从这个定义开始

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle$$

将公式(12)中的前两个 $f_q(\mathbf{x}_m, m) = (\mathbf{W}_q \mathbf{x}_m) e^{im\theta}$, $f_k(\mathbf{x}_n, n) = (\mathbf{W}_k \mathbf{x}_n) e^{in\theta}$ 代入上式，注意，这两个都是复数，但是为了叙述简洁，下式中的内积，将这两项视为向量。

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \langle (\mathbf{W}_q \mathbf{x}_m) e^{im\theta}, (\mathbf{W}_k \mathbf{x}_n) e^{in\theta} \rangle$$

参考前面的例子， $\langle \vec{z}_1, \vec{z}_2 \rangle = (a_1 \cdot a_2 + b_1 \cdot b_2) = \text{Re}[z_1 \cdot (z_2)^*]$

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m) \cdot e^{im\theta} \cdot ((\mathbf{W}_k \mathbf{x}_n) \cdot e^{in\theta})^*]$$

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m) \cdot e^{im\theta} \cdot (e^{in\theta})^* \cdot (\mathbf{W}_k \mathbf{x}_n)^*]$$

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m) \cdot e^{im\theta} \cdot e^{i(-n\theta)} \cdot (\mathbf{W}_k \mathbf{x}_n)^*]$$

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m) \cdot e^{i(m-n)\theta} \cdot (\mathbf{W}_k \mathbf{x}_n)^*]$$

$$g(\mathbf{x}_m, \mathbf{x}_n, m-n) = \text{Re}[(\mathbf{W}_q \mathbf{x}_m) \cdot (\mathbf{W}_k \mathbf{x}_n)^* \cdot e^{i(m-n)\theta}]$$

假设有复数 z_1 , z_2 , $\begin{cases} z_1 = a_1 + i \cdot b_1 \\ z_2 = a_2 + i \cdot b_2 \end{cases}$, 则有 $(z_1 \cdot z_2)^* = (z_1)^* \cdot (z_2)^*$, 并且复数乘法满足交换律

$$\begin{aligned}
(z_1 \cdot z_2)^* &= ((a_1 + i \cdot b_1)(a_2 + i \cdot b_2))^* \\
&= ((a_1 \cdot a_2 - b_1 \cdot b_2) + i \cdot (a_1 b_2 + a_2 b_1))^* \\
&= (a_1 \cdot a_2 - b_1 \cdot b_2) - i \cdot (a_1 b_2 + a_2 b_1)
\end{aligned}$$

$$\begin{aligned}
(z_1)^* \cdot (z_2)^* &= (a_1 - i \cdot b_1)(a_2 - i \cdot b_2) \\
&= (a_1 \cdot a_2 - b_1 \cdot b_2) - i \cdot (a_1 b_2 + a_2 b_1)
\end{aligned}$$

Similarly, g can be viewed as a matrix and thus enables the solution of formulation in Section (3.1) under the 2D case.

Specifically, incorporating the relative position embedding is straightforward: simply **rotate** the affine-transformed word **embedding vector** by amount of **angle** multiples of its position index and thus interprets the intuition behind **Rotary Position Embedding**.

3.2.2 General form

In order to generalize our results in 2D to any $\mathbf{x}_i \in \mathbb{R}^d$ where d is even, we divide the d -dimension space into $d/2$ sub-spaces and combine them in the merit of the linearity of the inner product, turning $f_{\{q,k\}}$ into:

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \mathbf{R}_{\Theta, m}^d \mathbf{W}_{\{q,k\}} \mathbf{x}_m$$

(14)

Where

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}$$

(15)

is the rotary matrix with pre-defined parameters

$$\Theta = \left\{ \theta_i = 10000^{-\frac{2(i-1)}{d}}, \quad i \in \left[1, 2, \dots, \frac{d}{2} \right] \right\}$$

解释：将 d 维向量的所有分量两两分组，相邻的分量成为一组，如果 $d = 512$ ，则有 256 组。每一组的两个分量，就相当于是在二维复平面上的点，对其执行旋转操作。

所以大旋转矩阵 $\mathbf{R}_{\Theta, m}^d$ 就是一个分块对角阵，每个小块为一个二维旋转阵 $\begin{pmatrix} \cos m\theta_i & -\sin m\theta_i \\ \sin m\theta_i & \cos m\theta_i \end{pmatrix}$ 。

θ_i 的取值方法类似于 2017 transformer。

A graphic illustration of RoPE is shown in Figure (1).

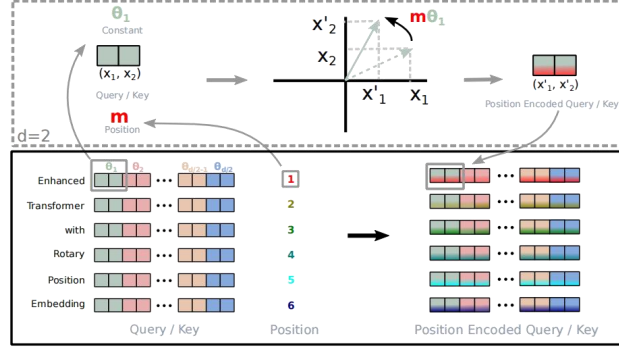


Figure 1: Implementation of Rotary Position Embedding(RoPE).

Applying our RoPE to self-attention in Equation (2), we obtain:

$$q_m^T k_n = (R_{\theta,m}^d W_q x_m)^T (R_{\theta,n}^d W_k x_n) = (W_q x_m)^T R_{\theta,n-m}^d (W_k x_n) \quad (16)$$

where $R_{\theta,n-m}^d = (R_{\theta,m}^d)^T (R_{\theta,n}^d)$.

Note that R_{θ}^d is an orthogonal matrix, which ensures stability during the process of encoding position information.

解释：公式(14)和公式(16)重写一下，

$$\begin{aligned} f_q(x_m, m) &= R_{\theta,m}^d (W_q x_m) \\ f_k(x_n, n) &= R_{\theta,n}^d (W_k x_n) \\ q_m^T k_n &= (R_{\theta,m}^d W_q x_m)^T (R_{\theta,n}^d W_k x_n) = (W_q x_m)^T R_{\theta,n-m}^d (W_k x_n) \end{aligned} \quad (16-1)$$

解释：公式(16-1)的前两个公式表明，位置信息嵌入是独立进行的，即绝对位置信息嵌入，

公式 $f_q(x_m, m) = R_{\theta,m}^d (W_q x_m)$ 表示位置 m 的信息嵌入 x_m 时，与其他位置 n 无关，
公式 $f_k(x_n, n) = R_{\theta,n}^d (W_k x_n)$ 表示位置 n 的信息嵌入 x_n 时，也与其他位置 m 无关，
而且这种嵌入可以拆分为与位置信息无关的投影 $(W_q x_m)$ 、 $(W_k x_n)$ 以及单纯的绝对位置信息嵌入算子（旋转矩阵） $R_{\theta,m}^d$ 、 $R_{\theta,n}^d$ ，旋转矩阵只包含绝对位置信息 m 或 n ，以及旋转参数组 θ ，

公式(16-1)的第三个公式表明，嵌入位置信息后，求相关度（attention）的表达式 $q_m^T k_n$ 可以拆分为两个与位置信息无关的投影 $(W_q x_m)^T$ 和 $(W_k x_n)$ ，以及单纯的相对位置信息嵌入算子（旋转矩阵） $R_{\theta,n-m}^d$ ，这个旋转矩阵只包含相对位置信息 $(n - m)$ ，以及旋转参数组 θ ，

解释：运算模块拆分表示的意义在于，从反向传播学习算法的角度，一个运算模块，能够按照信息来源进行拆分，则反向学习的路径就会很清晰，对模型设计和改进都很重要。

解释：公式(16)，

$$\begin{aligned}
\mathbf{q}_m^T \mathbf{k}_n &= (\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^T (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = (\mathbf{W}_q \mathbf{x}_m)^T (\mathbf{R}_{\Theta, m}^d)^T \mathbf{R}_{\Theta, n}^d (\mathbf{W}_k \mathbf{x}_n) \\
(\mathbf{R}_{\Theta, m}^d)^T \mathbf{R}_{\Theta, n}^d &= \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix}^T \\
&\quad \begin{pmatrix} \cos n\theta_1 & -\sin n\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ \sin n\theta_1 & \cos n\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \cos n\theta_2 & -\sin n\theta_2 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \sin n\theta_2 & \cos n\theta_2 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \cos n\theta_{d/2} & -\sin n\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \sin n\theta_{d/2} & \cos n\theta_{d/2} \end{pmatrix} \\
(\mathbf{R}_{\Theta, m}^d)^T \mathbf{R}_{\Theta, n}^d &= \begin{pmatrix} \cos m\theta_1 & \sin m\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ -\sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & \sin m\theta_2 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & -\sin m\theta_2 & \cos m\theta_2 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \cos m\theta_{d/2} & \sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \cdots & -\sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix} \\
&\quad \begin{pmatrix} \cos n\theta_1 & -\sin n\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ \sin n\theta_1 & \cos n\theta_1 & 0 & 0 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \cos n\theta_2 & -\sin n\theta_2 & \cdots & \cdots & 0 & 0 \\ 0 & 0 & \sin n\theta_2 & \cos n\theta_2 & \cdots & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \cos n\theta_{d/2} & -\sin n\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \cdots & \sin n\theta_{d/2} & \cos n\theta_{d/2} \end{pmatrix}
\end{aligned}$$

实际上是分块对角阵的矩阵乘法，只需要对相应子块矩阵做乘法即可，比如第一个子块的乘法及结果如下：

$$\begin{aligned}
&\begin{pmatrix} \cos m\theta_1 & \sin m\theta_1 \\ -\sin m\theta_1 & \cos m\theta_1 \end{pmatrix} \begin{pmatrix} \cos n\theta_1 & -\sin n\theta_1 \\ \sin n\theta_1 & \cos n\theta_1 \end{pmatrix} \\
&= \begin{pmatrix} (\cos m\theta_1 \cos n\theta_1 + \sin m\theta_1 \sin n\theta_1) & (-\cos m\theta_1 \sin n\theta_1 + \sin m\theta_1 \cos n\theta_1) \\ (-\sin m\theta_1 \cos n\theta_1 + \cos m\theta_1 \sin n\theta_1) & (\sin m\theta_1 \sin n\theta_1 + \cos m\theta_1 \cos n\theta_1) \end{pmatrix} \\
&= \begin{pmatrix} \cos (n-m)\theta_1 & -\sin (n-m)\theta_1 \\ \sin (n-m)\theta_1 & \cos (n-m)\theta_1 \end{pmatrix}
\end{aligned}$$

得到

$$\mathbf{R}_{\Theta, n-m}^d = (\mathbf{R}_{\Theta, m}^d)^T (\mathbf{R}_{\Theta, n}^d)$$

In addition, due to the sparsity of $\mathbf{R}_{\Theta}^d$, applying matrix multiplication directly as in Equation (16) is not computationally efficient; we provide another realization in theoretical explanation.

In contrast to the additive nature of position embedding method adopted in the previous works, i.e., Equations (3) to (10), our approach is multiplicative.

Moreover, RoPE naturally incorporates relative position information through rotation matrix product instead of altering terms in the expanded formulation of additive position encoding when applied with self-attention.

3.3 Properties of RoPE

Long-term decay: Following Vaswani et al. [2017], we set $\theta_i = 10000^{\frac{-2i}{d}}$.

One can prove that this setting provides a long-term decay property (refer to Section (3.4.3) for more details), which means the inner-product will decay when the relative position increase.

This property coincides with the intuition that a pair of tokens with a long relative distance should have less connection.

RoPE with linear attention: The self-attention can be rewritten in a more general form.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n)} \quad (17)$$

The original self-attention chooses $\text{sim}(\mathbf{q}_m, \mathbf{k}_n) = \exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_n}{\sqrt{d}}\right)$.

回顾公式(2),

$$a_{m,n} = \frac{\exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_n}{\sqrt{d}}\right)}{\sum_{j=1}^N \exp\left(\frac{\mathbf{q}_m^T \mathbf{k}_j}{\sqrt{d}}\right)} \quad \begin{cases} m = 1, \dots, N \\ n = 1, \dots, N \end{cases} \quad \mathbf{o}_m = \sum_{n=1}^N a_{m,n} \mathbf{v}_n \quad m = 1, \dots, N$$

公式(17)与公式(2)是一致的。

Note that the original self-attention should compute the inner product of query and key for every pair of tokens, which has a quadratic complexity $O(N^2)$.

Follow Katharopoulos et al. [2020], the **linear attentions** reformulate Equation (17) as

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \phi(\mathbf{q}_m)^T \varphi(\mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \phi(\mathbf{q}_m)^T \varphi(\mathbf{k}_n)} \quad (18)$$

where $\phi(\cdot)$, $\varphi(\cdot)$ are usually non-negative functions.

在公式(17)的基础上，提出更一般形式的 **linear attentions**,

The authors of Katharopoulos et al. [2020] (参考 2006.16236) have proposed $\phi(x) = \varphi(x) = \text{elu}(x) + 1$ and first computed the multiplication between keys and values using the associative property of matrix multiplication.

$\text{elu}(x)$ 指数线性单元, exponential linear unit, 为一种激活函数,

$$\text{elu}(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases} \quad \alpha = 1 \text{ or other}$$

A softmax function is used in Shen et al. [2021] (参考 1812.01243) to normalize queries and keys separately before the inner product, which is equivalent to $\phi(\mathbf{q}_i) = \text{softmax}(\mathbf{q}_i)$ and $\phi(\mathbf{k}_j) = \exp(\mathbf{k}_j)$.

For more details about linear attention, we encourage readers to refer to original papers.

In this section, we focus on discussing incorporating RoPE with Equation (18).

本文在公式(18)的基础上，讨论 RoPE 的性质。

Since RoPE injects position information by rotation, which **keeps the norm** of hidden representations unchanged, we can combine RoPE with linear attention by multiplying the rotation matrix with the outputs of the non-negative functions.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \left(\mathbf{R}_{\Theta, m}^d \phi(\mathbf{q}_m) \right)^T \left(\mathbf{R}_{\Theta, n}^d \varphi(\mathbf{k}_n) \right) \mathbf{v}_n}{\sum_{n=1}^N \phi(\mathbf{q}_m)^T \varphi(\mathbf{k}_n)}$$

(19)

It is noteworthy that we **keep the denominator unchanged** to avoid the risk of dividing zero, and the summation in the numerator could contain negative terms.

公式(19)中的分母部分不做旋转，只在分子部分做旋转。

Although the **weights** for each value $\mathbf{v}_i$ in Equation (19) are **not strictly probabilistic normalized**, we kindly argue that the computation can still model the importance of values.

3.4 Theoretical Explanation

3.4.1 Derivation of RoPE under 2D

Under the case of $d = 2$, we consider two-word embedding vectors $\mathbf{x}_q$, $\mathbf{x}_k$ corresponds to query and key and their position m and n , respectively.

考虑简化的二维情况，嵌入向量 $\mathbf{x}_q$, $\mathbf{x}_k$ 表示 query 和 key，位置分别为 m 和 n

According to eq. (1), their position-encoded counterparts are:

$$\begin{aligned}\mathbf{q}_m &= f_q(\mathbf{x}_q, m) \\ \mathbf{k}_n &= f_k(\mathbf{x}_k, n)\end{aligned}$$

(20)

where the subscripts of $\mathbf{q}_m$ and $\mathbf{k}_n$ indicate the encoded positions information.

公式(20)先写出位置嵌入函数 f_q, f_k 的一般形式。

Assume that there exists a function g that defines the inner product between vectors produced by $f_{\{q,k\}}$:

$$\mathbf{q}_m^T \mathbf{k}_n = \langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, m - n)$$

(21)

公式(21)说明位置嵌入的目的，是为了计算内积 $\mathbf{q}_m^T \mathbf{k}_n$,

we further require below initial condition to be satisfied:

$$\begin{aligned}\mathbf{q}_m &= f_q(\mathbf{x}_q, 0) \\ \mathbf{k}_n &= f_k(\mathbf{x}_k, 0)\end{aligned}$$

(22)

which can be read as the vectors with empty position information encoded.

公式(22)提出了平凡条件，即不嵌入位置的情况下，位置嵌入函数 f_q, f_k 应该满足的条件。

Given these settings, we attempt to **find a solution** of f_q, f_k .

解释：这里提出了条件，下面寻找符合条件的位置嵌入函数 f_q, f_k

First, we take advantage of the geometric meaning of vector in 2D and its complex counter part, decompose functions in Equations (20) and (21) into:

$$\begin{aligned}f_q(\mathbf{x}_q, m) &= R_q(\mathbf{x}_q, m) e^{i\Theta_q(\mathbf{x}_q, m)} \\ f_k(\mathbf{x}_k, n) &= R_k(\mathbf{x}_k, n) e^{i\Theta_k(\mathbf{x}_k, n)} \\ g(\mathbf{x}_q, \mathbf{x}_k, n - m) &= R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) e^{i\Theta_g(\mathbf{x}_q, \mathbf{x}_k, n - m)}\end{aligned}$$

(23)

where R_f, R_g and Θ_f, Θ_g are the radical and angular components for $f_{\{q,k\}}$ and g , respectively. radical and angular 复数的模和角度

解释：这里的主要思路是，在二维复平面上做直观描述，认为位置嵌入函数的操作，相当于是对二维复平面上复数点（对应二维向量）进行位置变换（复数点移位），而复数点移位又可以分解为复数点的模和角度的变换。

Plug them into Equation (21), we get the relation:

$$\begin{aligned}R_q(\mathbf{x}_q, m) R_k(\mathbf{x}_k, n) &= R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) \\ \Theta_k(\mathbf{x}_k, n) - \Theta_q(\mathbf{x}_q, m) &= \Theta_g(\mathbf{x}_q, \mathbf{x}_k, n - m)\end{aligned}$$

(24)

公式(24)表明，模和角度的变换可以分开来表达。

with the corresponding initial condition as:

$$\mathbf{q} = \|\mathbf{q}\| e^{i\theta_q} = R_q(\mathbf{x}_q, 0) e^{i\Theta_q(\mathbf{x}_q, 0)}$$

$$\mathbf{k} = \|\mathbf{k}\|e^{i\theta_k} = R_k(\mathbf{x}_k, 0)e^{i\Theta_k(\mathbf{x}_k, 0)} \quad (25)$$

where $\|\mathbf{q}\|$, $\|\mathbf{k}\|$ and θ_q , θ_k are the radial and angular part of $\mathbf{q}$ and $\mathbf{k}$ on the 2D plane. 公式(25), 在模和角度的变换分开表达后, 重写了公式(22)提出了平凡条件 (初始化条件),

Next, we set $m = n$ in Equation (24) and take into account initial conditions in Equation (25):

$$R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, m) = R_g(\mathbf{x}_q, \mathbf{x}_k, 0) = R_q(\mathbf{x}_q, 0)R_k(\mathbf{x}_k, 0) = \|\mathbf{q}\| \cdot \|\mathbf{k}\| \quad (26a)$$

$$\Theta_k(\mathbf{x}_k, m) - \Theta_q(\mathbf{x}_q, m) = \Theta_g(\mathbf{x}_q, \mathbf{x}_k, 0) = \Theta_k(\mathbf{x}_k, 0) - \Theta_q(\mathbf{x}_q, 0) = \theta_k - \theta_q \quad (26b)$$

公式(26), 在 $m = n$ 的条件下, 模和角度的变换结果,

On one hand, from, a straightforward solution of R_f could be formed from Equation (26a) :

$$\begin{aligned} R_q(\mathbf{x}_q, m) &= R_q(\mathbf{x}_q, 0) = \|\mathbf{q}\| \\ R_k(\mathbf{x}_k, n) &= R_k(\mathbf{x}_k, 0) = \|\mathbf{k}\| \\ R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) &= R_g(\mathbf{x}_q, \mathbf{x}_k, 0) = \|\mathbf{q}\| \cdot \|\mathbf{k}\| \end{aligned} \quad (27)$$

which interprets the radial functions R_q , R_k and R_g are independent from the position information.

公式(27)表明, 模的变换是独立于位置信息的, 也就是说, 位置嵌入操作, 只改变复数点 (向量) 的角度。

On the other hand, as can be noticed in Equation (26b), 将公式(26b)写为如下形式,

$$\Theta_q(\mathbf{x}_q, m) - \theta_q = \Theta_k(\mathbf{x}_k, m) - \theta_k$$

此式的左右两边, 左边是 query, 右边是 key, 但是形式上是一样的, 即角度变换对 query 和 key 是同类函数, 只是参数 θ_q , θ_k 不同,

indicates that the angular functions does not dependent on query and key, we set them to

$$\Theta_f = \Theta_q = \Theta_k$$

and term

$$\Theta_f(\mathbf{x}_{\{q,k\}}, m) - \theta_{\{q,k\}}$$

is a function of position m and is independent of word embedding $\mathbf{x}_{\{q,k\}}$, we denote it as $\phi(m)$, yielding:

$$\Theta_f(\mathbf{x}_{\{q,k\}}, m) = \phi(m) + \theta_{\{q,k\}} \quad (28)$$

公式(28)给出了角度变换的一般形式, 对 query 和 key 都适用, 但此时, $\phi(m)$ 的具体形式还不能确定,

Further, by plugging $n = m + 1$ to Equation (24) and consider the above equation, we can get:

$$\phi(m+1) - \phi(m) = \Theta_g(\mathbf{x}_q, \mathbf{x}_k, 1) + \theta_q - \theta_k$$

(29)

公式(29)是一种 $\phi(m)$ 的递推形式,

Since RHS is a constant irrelevant to m , $\phi(m)$ with continuous integer inputs produce an arithmetic progression:

$$\phi(m) = m\theta + \gamma$$

(30)

where $\theta, \gamma \in \mathbb{R}$ are constants and θ is non-zero.

公式(30), 根据公式(29)的递推形式, 认为 $\phi(m)$ 是一个线性函数, $\theta, \gamma \in \mathbb{R}$ 是其参数。

To summarize our solutions from Equations (27) to (30):

$$f_q(\mathbf{x}_q, m) = \|\mathbf{q}\|e^{i(\theta_q+m\theta+\gamma)} = \mathbf{q}e^{i(m\theta+\gamma)}$$

$$f_k(\mathbf{x}_k, n) = \|\mathbf{k}\|e^{i(\theta_k+n\theta+\gamma)} = \mathbf{k}e^{i(n\theta+\gamma)}$$

(31)

注释: 原文此公式中间项有误 ($e^{i\theta_q+m\theta+\gamma}$, $e^{i\theta_k+n\theta+\gamma}$)

这里用到了公式(25)中的 $\begin{cases} \mathbf{q} = \|\mathbf{q}\|e^{i\theta_q} \\ \mathbf{k} = \|\mathbf{k}\|e^{i\theta_k} \end{cases}$

公式(31)就是结论, (对二维情况而言) 位置嵌入变换的一般形式, 就是**模不变, 角度线性变化**。

Note that we do not apply any constrains to f_q and f_k of Equation (22), thus $f_q(\mathbf{x}_m, 0)$ and $f_k(\mathbf{x}_n, 0)$ are left to choose freely.

To make our results comparable to Equation (3), we define:

$$\mathbf{q} = f_q(\mathbf{x}_m, 0) = \mathbf{W}_q \mathbf{x}_m$$

$$\mathbf{k} = f_k(\mathbf{x}_n, 0) = \mathbf{W}_k \mathbf{x}_n$$

(32)

公式(32), 不嵌入位置的情况下 ($m=0, n=0$), 只做投影变换 (线性变换),

Then, we simply set $\gamma = 0$ in Equation (31) of the final solution:

$$f_q(\mathbf{x}_m, m) = (\mathbf{W}_q \mathbf{x}_m) e^{im\theta}$$

$$f_k(\mathbf{x}_n, n) = (\mathbf{W}_k \mathbf{x}_n) e^{in\theta}$$

(33)

公式(33), 嵌入位置的情况下 ($m \neq 0, n \neq 0$), 先做投影, 再做旋转。

3.4.2 Computational efficient realization of rotary matrix multiplication

Taking the advantage of the sparsity of $\mathbf{R}_{\theta, m}^d$ in Equation (15), a more computational efficient realization of a multiplication of $\mathbf{R}_{\theta}^d$ and $\mathbf{x} \in \mathbb{R}^d$ is:

$$\mathbf{R}_{\Theta,m}^d \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \cos m\theta_2 \\ \cos m\theta_2 \\ \vdots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_1 \\ -x_4 \\ x_3 \\ \vdots \\ -x_d \\ x_{d-1} \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \sin m\theta_2 \\ \sin m\theta_2 \\ \vdots \\ \sin m\theta_{d/2} \\ \sin m\theta_{d/2} \end{pmatrix}$$